Algorithm

Final Exam, June 24, 2021

5:30-8:30 pm

1. (15%) Please answer the following questions:

   (a) Illustrate the operation of Radix-Sort on the following numbers: 268, 356, 793, 254, 257, 399, 358, 921, 331, 536.

   (b) Show that if a node in a binary search tree has two children, then its successor has no left child.

   (c) For the set of keys $\{5, 22, 13, 15, 27, 33, 11\}$, draw binary search trees of height 2, 3, 4, 5 and 6.

2. (15%) Given a sequence $X = \langle x_1, x_2, \ldots, x_m \rangle$, the longest increasing subsequence problem is to find an increasing subsequence of X with the longest length.

   (a) Find a longest increasing subsequence of $\langle 6, 35, 24, 7, 8, 26, 21, 34 \rangle$.

   (b) Describe an algorithm that solves the longest increasing subsequence problem in $O(n^2)$ time.

3. (20%) Consider inserting the keys $5, 22, 13, 15, 27, 33, 11, 21$ into a hash table of length $m = 13$ using open addressing with the auxiliary hash function $h'(k) = k$. Show the results after the insertion of each number using the following methods.

   (a) Linear hashing.

   (b) Quadratic probing with $c_1 = 1$ and $c_2 = 3$.

   (c) Double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \mod (m-1))$.

   (d) Chaining.

4. (15%) Medians and Order Statistics.

   (a) In the algorithm Select, the input elements are divided into groups of 5. Will the algorithm work in linear time if they are divided into groups of 9?

   (b) Show how Quicksort can be made to run in $O(n \lg n)$ time in the worst case.

5. (15%) Red-Black Trees.

   (a) Show the red-black trees that result after successively inserting the keys $5, 22, 13, 15, 27, 33, 11, 21$ into an initially empty red-black tree.

   (b) What is the largest possible number of internal nodes in a red-black tree with black height k? What is the smallest possible number?

6. (15%)Minimizing lateness problem.

- Single resource processes one job at a time.
- Job i requires t_i units of processing time and is due at time d_i.
- If i starts at time s_i, it finishes at time $f_i = s_i + t_i$.
- Lateness: $\ell_i = \max\{0, f_i - d_i\}$.
- Goal: schedule all jobs to minimize maximum lateness $L = \max\{\ell_i \mid 1 \leq i \leq n\}$.

i	1	2	3	4	5	6	7
t_i	3	7	4	1	2	7	5
d_i	5	9	8	8	9	13	11

7. (20%) Alice is playing a game where she controls a character to walk on a $m \times n$ table. The character starts at the bottom-left corner (coordinate $(1,1)$) and wants to go to the top-right corner (coordinate (m,n)). The character can only move to the right (from (i,j) to $(i,j+1)$) or up (from (i,j) to $(i+1,j)$). Each square of the table has a non-negative value $v[i,j]$. The goal is to find a path from the starting point to the end point that maximizes the sum of $v[i,j]$ for all the squares (i,j) on the path. See the left figure below, the arrows show the optimal path with the maximum total sum of all its values.

   (a) Define the subproblems, write the recursive formula, and specify the base cases.

   (b) Design an algorithm that outputs the maximum possible value. Your algorithm should run in time $O(mn)$.

   (c) Suppose the character learned two new moves that allows it to move from (i,j) to $(i+2, j-1)$ and $(i-1, j+2)$ (See the right figure below). Explain how to modify the recursive formula and the algorithm.